

```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from matplotlib.lines import Line2D
```

— 1. CARREGAR DADOS

```
df = pd.read_excel("703pacientes.xlsx")
```

— 2. PREPARAR VARIÁVEIS

```
df["Data de Entrada"] = pd.to_datetime(df["Data de Entrada"], dayfirst=True)
```

— 3. AGREGAR POR MÊS

```
mensal = df.groupby(df["Data de Entrada"].dt.to_period("M")).agg(
    n_internacoes=("Data de Entrada", "count"),
    n_obitos=("Óbito", "sum")
).reset_index()
mensal["n_altas"] = mensal["n_internacoes"] - mensal["n_obitos"]

mensal.rename(columns={"Data de Entrada": "mes"}, inplace=True)
mensal["mes_dt"] = mensal["mes"].dt.to_timestamp()
```

— 4. DEFINIR PERÍODOS DAS ONDAS

```
ondas = [
    {"label": "1st Wave", "inicio": "2020-01-01", "fim": "2020-09-30", "cor": "#7b5ea7"},
    {"label": "2nd Wave", "inicio": "2020-10-01", "fim": "2021-05-31", "cor": "#c0823a"},
    {"label": "3rd Wave", "inicio": "2021-07-01", "fim": "2021-11-30", "cor": "#b03060"},
    {"label": "4th Wave", "inicio": "2021-12-01", "fim": "2022-12-30", "cor": "#5b8db8"},
]
```

— 5. PLOTAR

```
fig, ax = plt.subplots(figsize=(14, 6))
```

```
# Fundo colorido por onda
```

```
for onda in ondas:
```

```
    ax.axvspan(
        pd.to_datetime(onda["inicio"]),
        pd.to_datetime(onda["fim"]),
        alpha=0.3, color=onda["cor"], label=onda["label"]
    )
```

```
# Area chart — internações
```

```
ax.fill_between(mensal["mes_dt"], mensal["n_internacoes"],
               alpha=0.6, color="#2980b9", label="Internações")
ax.plot(mensal["mes_dt"], mensal["n_internacoes"],
       color="#2980b9", linewidth=2.8)
```

```
# Area chart — altas hospitalares
```

```
ax.fill_between(mensal["mes_dt"], mensal["n_altas"],
               alpha=0.6, color="#27ae60", label="Altas")
ax.plot(mensal["mes_dt"], mensal["n_altas"],
       color="#1e8449", linewidth=2.8)
```

```
# Area chart — óbitos
```

```
ax.fill_between(mensal["mes_dt"], mensal["n_obitos"],
               alpha=0.7, color="#e74c3c", label="Óbitos")
```

```

ax.plot(mensal["mes_dt"], mensal["n_obitos"],
        color="#c0392b", linewidth=2.8)

# Linha tracejada — início da vacinação
vacinacao = pd.to_datetime("2021-01-19")
ax.axvline(x=vacinacao, color="black", linestyle="--", linewidth=2.8)
ax.text(vacinacao + pd.Timedelta(days=5), ax.get_ylim()[1] * 0.99,
        "Start of COVID-19 Vaccination\n(19/01/2021)",
        fontsize=12, color="black", va="top")

# — 6. FORMATAÇÃO


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ax.set_xlim(pd.to_datetime("2020-01-01"), pd.to_datetime("2022-12-31"))
ax.xaxis.set_major_formatter(mdates.DateFormatter("%b/%Y"))
ax.xaxis.set_major_locator(mdates.MonthLocator(interval=3))
plt.xticks(rotation=45, ha="right", fontsize=12)
plt.yticks(fontsize=12)

ax.set_xlabel("Period", fontsize=15, fontweight="bold")
ax.set_ylabel("Number of Patients", fontsize=15, fontweight="bold")
ax.set_title("Temporal Trends in Hospitalizations and Deaths Across COVID-19 Waves\n(Jan/2020 –
Dez/2022)",
            fontsize=22, fontweight="bold", pad=25)

# Legenda dentro do gráfico
legend_elementos = [
    Line2D([0], [0], color="#2980b9", lw=3, label="Hospitalizations (total)",
    Line2D([0], [0], color="#27ae60", lw=3, label="Discharge"),
    Line2D([0], [0], color="#e74c3c", lw=3, label="Death"),
    Line2D([0], [0], color="black", lw=1.8, linestyle="--", label="Start of Vaccination"),
    plt.Rectangle((0, 0), 1, 1, fc="#7b5ea7", alpha=0.5, label="1st Wave"),
    plt.Rectangle((0, 0), 1, 1, fc="#c0823a", alpha=0.5, label="2nd Wave"),
    plt.Rectangle((0, 0), 1, 1, fc="#b03060", alpha=0.5, label="3rd Wave"),
    plt.Rectangle((0, 0), 1, 1, fc="#5b8db8", alpha=0.5, label="4th Wave"),
]
ax.legend(handles=legend_elementos, loc="upper right", fontsize=10,
        framealpha=0.9, bbox_to_anchor=(1.001, 1.01))

ax.spines["top"].set_visible(False)
ax.spines["right"].set_visible(False)

# — 8. ANOTAÇÕES ESTATÍSTICAS POR ONDA


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stats_ondas = [
    {"x": "2020-05-01", "label": "1st Wave\n(ref.)"},
    {"x": "2020-11-15", "label": "2nd Wave\nOR=0.67 [0.45–0.98]\np=0.041"},
    {"x": "2021-09-01", "label": "3rd Wave\nOR=0.64 [0.40–1.03]\np=0.068"},
    {"x": "2022-04-01", "label": "4th Wave\nOR=0.14 [0.09–0.23]\np<0.001"},
]

for s in stats_ondas:
    ax.text(
        pd.to_datetime(s["x"]), 0.85,
        s["label"],
        transform=ax.get_xaxis_transform(),
        fontsize=8, color="#333333",
        ha="center", va="top",
        bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="gray",
            alpha=0.59, linewidth=0.9)
    )

```

```
ax.text(
    0.13, 0.99,
    " $\chi^2=72.88$ , df=3, p<0.001",
    transform=ax.transAxes,
    fontsize=9, color="#555555",
    ha="right", va="bottom",
    bbox=dict(boxstyle="round,pad=0.3", fc="white", ec="gray",
        alpha=0.99, linewidth=0.6)
)
plt.tight_layout()
```

```
# — 7. SALVAR
```

```
plt.savefig("grafico_tendencia_covid.png", dpi=300, bbox_inches="tight")
plt.savefig("grafico_tendencia_covid.pdf", bbox_inches="tight")
plt.show()
print("Gráfico salvo como PNG e PDF com sucesso.")
```